Databases in AWS

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Choosing the Right Database

- We have a lot of managed databases on AWS to choose from
- Questions to choose the right database based on your architecture:
  - Read-heavy, write-heavy, or balanced workload? Throughput needs? Will it change, does it need to scale or fluctuate during the day?
  - How much data to store and for how long? Will it grow? Average object size? How are they accessed?
  - Data durability? Source of truth for the data ?
  - Latency requirements? Concurrent users?
  - Data model? How will you query the data? Joins? Structured? Semi-Structured?
  - Strong schema? More flexibility? Reporting? Search? RDBMS / NoSQL?
  - License costs? Switch to Cloud Native DB such as Aurora?

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Database Types



- **RDBMS (= SQL / OLTP):** RDS, Aurora – great for joins
- **NoSQL database – no joins, no SQL :** DynamoDB (~JSON), ElastiCache (key / value pairs), Neptune (graphs), DocumentDB (for MongoDB), Keyspaces (for Apache Cassandra)
- **Object Store:** S3 (for big objects) / Glacier (for backups / archives)
- **Data Warehouse** (= SQL Analytics / BI): Redshift (OLAP), Athena, EMR
- **Search:** OpenSearch (JSON) – free text, unstructured searches
- **Graphs:** Amazon Neptune – displays relationships between data
- **Ledger**: Amazon Quantum Ledger Database
- **Time series**: Amazon Timestream

- Note: some databases are being discussed in the Data & Analytics section

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Amazon RDS – Summary



- Managed PostgreSQL / MySQL / Oracle / SQL Server / DB2 / MariaDB / Custom
- Provisioned RDS Instance Size and EBS Volume Type & Size
- Auto-scaling capability for Storage
- Support for Read Replicas and Multi AZ
- Security through IAM, Security Groups, KMS , SSL in transit
- Automated Backup with Point in time restore feature (up to 35 days)
- Manual DB Snapshot for longer-term recovery
- Managed and Scheduled maintenance (with downtime)
- Support for IAM Authentication, integration with Secrets Manager
- RDS Custom for access to and customize the underlying instance (Oracle & SQL Server)

- **Use case:** Store relational datasets (RDBMS / OLTP), perform SQL queries, transactions

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Amazon Aurora – Summary



- Compatible API for PostgreSQL / MySQL, separation of storage and compute
- Storage: data is stored in 6 replicas, across 3 AZ – highly available, self-healing, auto-scaling
- Compute: Cluster of DB Instance across multiple AZ, auto-scaling of Read Replicas
- Cluster: Custom endpoints for writer and reader DB instances
- Same security / monitoring / maintenance features as RDS
- Know the backup & restore options for Aurora
- **Aurora Serverless** – for unpredictable / intermittent workloads, no capacity planning
- **Aurora Global:** up to 16 DB Read Instances in each region, < 1 second storage replication
- **Aurora Machine Learning:** perform ML using SageMaker & Comprehend on Aurora
- **Aurora Database Cloning:** new cluster from existing one, faster than restoring a snapshot
- **Use case:** same as RDS, but with less maintenance / more flexibility / more performance / more features

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Amazon ElastiCache – Summary



- Managed Redis / Memcached (similar offering as RDS, but for caches)
- In-memory data store, sub-millisecond latency
- Select an ElastiCache instance type (e.g., cache.m6g.large)
- Support for Clustering (Redis) and Multi AZ, Read Replicas (sharding)
- Security through IAM, Security Groups, KMS, Redis Auth
- Backup / Snapshot / Point in time restore feature
- Managed and Scheduled maintenance
- **Requires some application code changes to be leveraged**

- **Use Case:** Key/Value store, Frequent reads, less writes, cache results for DB queries, store session data for websites, cannot use SQL.

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Amazon DynamoDB – Summary



- AWS proprietary technology, managed serverless NoSQL database, millisecond latency
- Capacity modes: provisioned capacity with optional auto-scaling or on-demand capacity
- Can replace ElastiCache as a key/value store (storing session data for example, using TTL feature)
- Highly Available, Multi AZ by default, Read and Writes are decoupled, transaction capability
- DAX cluster for read cache, microsecond read latency
- Security, authentication and authorization is done through IAM
- Event Processing: DynamoDB Streams to integrate with AWS Lambda, or Kinesis Data Streams
- Global Table feature: active-active setup
- Automated backups up to 35 days with PITR (restore to new table), or on-demand backups
- Export to S3 without using RCU within the PITR window, import from S3 without using WCU
- **Great to rapidly evolve schemas**

- **Use Case:** Serverless applications development (small documents 100s KB), distributed serverless cache

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Amazon S3 – Summary



- S3 is a… key / value store for objects
- Great for bigger objects, not so great for many small objects
- Serverless, scales infinitely, max object size is 50 TB, versioning capability
- **Tiers:** S3 Standard, S3 Infrequent Access, S3 Intelligent, S3 Glacier + lifecycle policy
- **Features:** Versioning, Encryption, Replication, MFA-Delete, Access Logs…
- **Security:** IAM, Bucket Policies, ACL, Access Points, Object Lambda, CORS, Object/Vault Lock
- **Encryption:** SSE-S3, SSE-KMS, SSE-C, client-side, TLS in transit, default encryption
- **Batch operations** on objects using S3 Batch, listing files using S3 Inventory
- **Performance:** Multi-part upload, S3 Transfer Acceleration, S3 Select
- **Automation:** S3 Event Notifications (SNS, SQS, Lambda, EventBridge)
- **Use Cases:** static files, key value store for big files, website hosting

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DocumentDB





- Aurora is an "AWS-implementation" of PostgreSQL / MySQL …
- **DocumentDB is the same for MongoDB (which is a NoSQL database)**

- MongoDB is used to store, query, and index JSON data
- Similar "deployment concepts" as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB

- Automatically scales to workloads with millions of requests per seconds

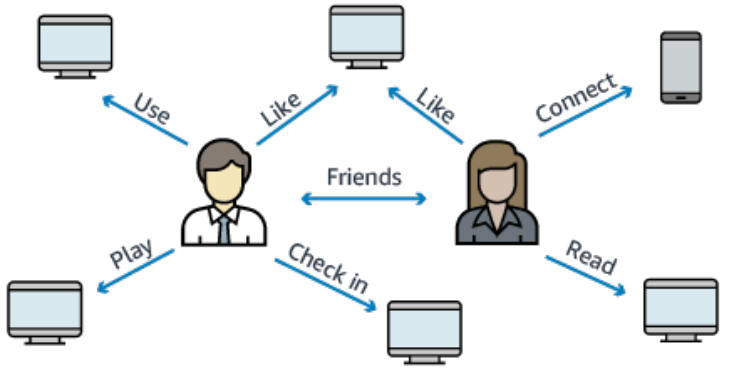
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Amazon Neptune



- Fully managed **graph** database
- A popular **graph dataset** would be a **social network**
  - Users have friends
  - Posts have comments
  - Comments have likes from users
  - Users share and like posts…
- Highly available across 3 AZ, with up to 15 read replicas
- Build and run applications working with highly connected datasets – optimized for these complex and hard queries
- Can store up to billions of relations and query the graph with milliseconds latency
- Highly available with replications across multiple AZs
- Great for knowledge graphs (Wikipedia), fraud detection, recommendation engines, social networking

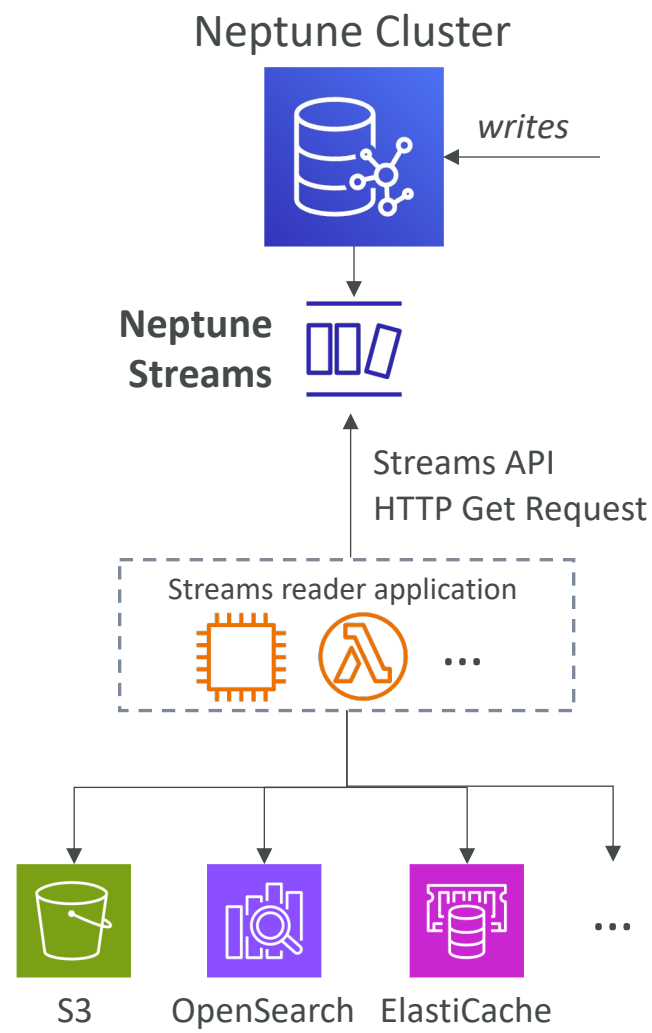


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Amazon Neptune – Streams

- **Real-time ordered** sequence of every change to your graph data
- Changes are available immediately after writing
- **No duplicates, strict order**
- Streams data is accessible in an **HTTP REST API**
- Use cases:
  - Send notifications when certain changes are made
  - Maintain your graph data synchronized in another data store (e.g., S3, OpenSearch, ElastiCache)
  - Replicate data across regions in Neptune



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Amazon Keyspaces (for Apache Cassandra)



- *Apache Cassandra is an open-source NoSQL distributed database*

- A managed Apache Cassandra-compatible database service
- Serverless, Scalable, highly available, fully managed by AWS
- Automatically scale tables up/down based on the application's traffic
- Tables are replicated 3 times across multiple AZ
- Using the Cassandra Query Language (CQL)
- Single-digit millisecond latency at any scale, 1000s of requests per second
- Capacity: On-demand mode or provisioned mode with auto-scaling
- Encryption, backup, Point-In-Time Recovery (PITR) up to 35 days

- Use cases: store IoT devices info, time-series data, …

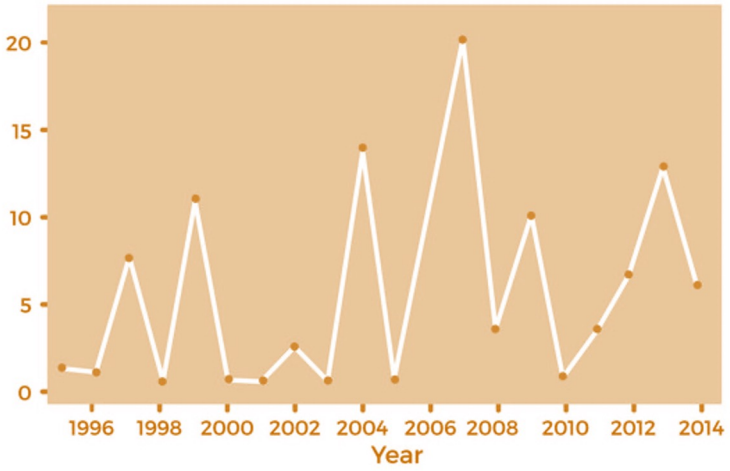
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Amazon Timestream



- Fully managed, fast, scalable, serverless **time series database**
- Automatically scales up/down to adjust capacity
- Store and analyze trillions of events per day
- 1000s times faster & 1/10th the cost of relational databases
- Scheduled queries, multi-measure records, SQL compatibility
- Data storage tiering: recent data kept in memory and historical data kept in a cost-optimized storage
- Built-in time series analytics functions (helps you identify patterns in your data in near real-time)
- Encryption in transit and at rest

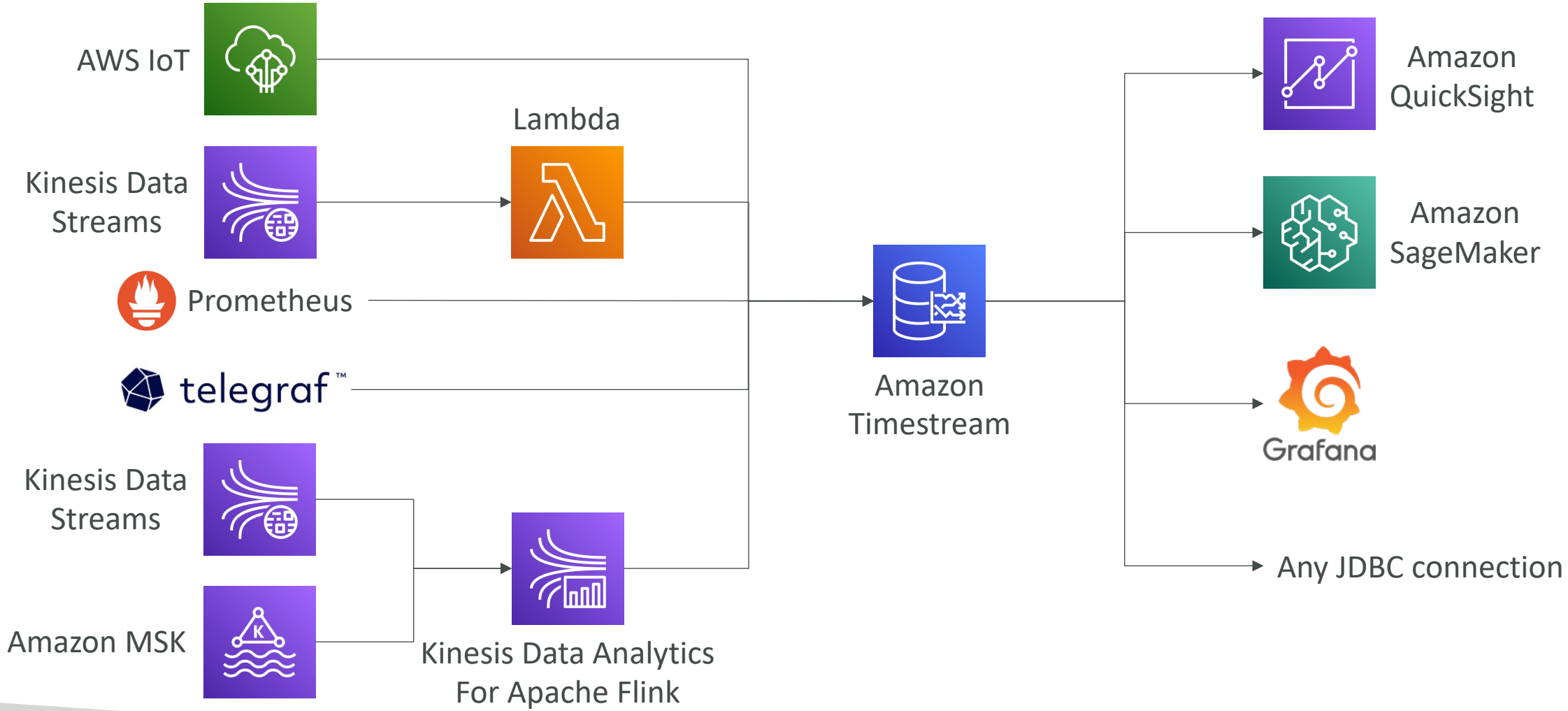


- Use cases: IoT apps, operational applications, real-time analytics, …

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Amazon Timestream – Architecture



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